

FLIPPER BLACK

Complete Schematic Roadmap

This guide walks you through building the entire Flipper Black schematic in KiCad, chip by chip, step by step. Every connection is explained — what connects to what, why, and what label to use. Follow every chip section in order. Do not skip ahead.

WARNING: Web-verified against Espressif Hardware Design Guidelines (April 2026), CC1101 datasheet (TI), and SX1276 datasheet (Semtech). Always cross-check pin numbers against your exact chip datasheet before placing wires.

Build order

1. ESP32-S3 — main MCU (start here)
2. W25Q128 — SPI flash (do immediately after MCU)
3. CC1101 — sub-GHz radio + RF matching network
4. SX1276 — LoRa transceiver + RF matching network
5. PN532 — NFC / RFID controller
6. MAX-M8Q — GPS + TLV70018 LDO + TXB0102 level shifter
7. Display + touch controller
8. IR TX / RX
9. Power system (TP4056 + DW01A + AP2112K + battery)
10. Buttons, test points, USB-C connector

The ESP32-S3 is the brain. Every other chip connects to it. You must finish this chip completely — every power pin, every decoupling cap, every strapping pin, the crystal, the flash SPI pins, the USB pins, and all net labels. Only then move to Chip 2.

1A — Power pins

The ESP32-S3 has multiple VDD pins. Each one needs its own 100nF decoupling capacitor placed directly on it. This is verified from the official Espressif Hardware Design Guidelines (April 2026).

Pin name	Pin number	What to connect	Cap required
VDD3P3_CPU	46	+3V3 power symbol	100nF to GND
VDD3P3_RTC	20	+3V3 power symbol	100nF to GND
VDD3P3 (analog)	2, 3	+3V3 power symbol	100nF + 1uF + 10uF to GND — RF transmit causes current spikes
VDDA	55, 56	+3V3 power symbol	100nF to GND each pin
VDD_SPI	29	+3V3 power symbol	100nF + 1uF to GND — powers W25Q128 flash
GND	All GND pins	GND symbol	No cap needed
Thermal pad (bottom)	EP	GND symbol	Connect to large GND pad with minimum 9 vias in PCB layout

CRITICAL: Add a 10uF capacitor on VDD3P3 (pins 2 and 3). When ESP32-S3 transmits WiFi signals it draws sudden current spikes that collapse the power rail. The 10uF bulk cap prevents this. Verified from Espressif guidelines.

1B — Crystal (40 MHz — mandatory)

ESP32-S3 requires an external 40 MHz crystal. This is not optional. Without it the chip will not boot. Connect between XTAL_P and XTAL_N pins.

ESP32-S3 pin	Connect to	Value	Notes
XTAL_P (XIN)	Crystal pin 1 + load cap C1 to GND	Crystal: 40MHz SMD 3225	Use NP0/C0G dielectric for load caps — NOT X7R or Y5V
XTAL_N (XOUT)	Crystal pin 2 + load cap C2 to GND	Crystal: 40MHz SMD 3225	Load cap value matches crystal datasheet CL spec (typically 10pF)
XTAL_P trace	Series inductor R4 between XIN and crystal	24nH inductor (initial value)	Reduces crystal harmonics affecting RF. Espressif recommends starting here.

TIP: Crystal load cap formula from Espressif: $CL = (C1 \times C2) / (C1 + C2) + C_{stray}$. If your crystal CL is 10pF and Cstray is ~2pF, use 22pF load caps. Check your crystal's datasheet CL value.

1C — Strapping pins (boot configuration)

Strapping pins are read at power-on to decide how the chip boots. Get these wrong and the chip won't start or will always enter download mode. These are verified from Espressif Hardware Design Guidelines.

GPIO	Net label	Connection	Why
GPIO0	BOOT_BTN	10k pull-up to +3V3. Tactile button to GND.	HIGH = normal boot. LOW = download/flash mode. Button lets you enter flash mode.
GPIO45	GPIO45_STRAP	10k pull-down to GND (resistor to GND).	Controls VDD_SPI voltage. LOW = 3.3V for W25Q128 flash (default needed).
GPIO46	GPIO46_STRAP	10k pull-down to GND (resistor to GND).	Must be LOW at boot. Do not leave floating.
CHIP_PU (EN)	EN_BTN	10k pull-up to +3V3. Tactile button to GND. RC filter: 1uF cap to GND on this pin.	Enables chip when HIGH. Button pulls LOW to reset. RC filter prevents false resets.

WARNING: Do not add high-value capacitors on GPIO0. A large cap on GPIO0 keeps it LOW too long at startup, causing the chip to enter download mode every boot. The RC filter goes on CHIP_PU (EN) only, not on GPIO0.

1D — SPI net labels (signal pins to other chips)

These GPIO pins connect to other chips via net labels. Place the label on the GPIO pin. The matching label goes on the destination chip pin. This is how chips talk to each other without drawing wires across the sheet.

GPIO	Net label	Connects to	Direction
GPIO11	SPI2_MOSI	CC1101 SI, PN532 MOSI (shared bus)	ESP32 sends data OUT
GPIO12	SPI2_SCK	CC1101 SCLK, PN532 SCK (shared bus)	ESP32 sends clock OUT
GPIO13	SPI2_MISO	CC1101 SO, PN532 MISO (shared bus)	ESP32 receives data IN
GPIO10	SPI2_CS_CC1101	CC1101 CSn only	ESP32 selects CC1101
GPIO9	SPI2_CS_PN532	PN532 NSS only	ESP32 selects PN532
GPIO35	SPI3_MOSI	SX1276 MOSI, W25Q128 DI (shared bus)	ESP32 sends data OUT
GPIO36	SPI3_SCK	SX1276 SCK, W25Q128 CLK (shared bus)	ESP32 sends clock OUT
GPIO37	SPI3_MISO	SX1276 MISO, W25Q128 DO (shared bus)	ESP32 receives data IN
GPIO34	SPI3_CS_SX1276	SX1276 NSS only	ESP32 selects SX1276
GPIO33	SPI3_CS_FLASH	W25Q128 /CS only	ESP32 selects flash
GPIO43	ESP_UART_TX	TXB0102 level shifter B2	GPS config commands OUT (add 100 ohm series resistor — boot conflict)
GPIO44	ESP_UART_RX	TXB0102 level shifter B1	GPS NMEA data IN
GPIO4	CC1101_GDO0	RF switch control + CC1101 GDO0	TX/RX state signal from CC1101

GPIO	Net label	Connects to	Direction
GPIO5	CC1101_GDO2	CC1101 GDO2	Packet interrupt from CC1101
GPIO21	SX1276_DIO0	SX1276 DIO0	TX done / RX done interrupt from SX1276
GPIO47	SX1276_RST	SX1276 RESET	ESP32 resets SX1276 — add 10k pull-up to +3V3
GPIO6	PN532_IRQ	PN532 IRQ	Card detected interrupt from PN532
GPIO1	I2C_SDA	Touch controller SDA	Bidirectional — add 4.7k pull-up to +3V3
GPIO2	I2C_SCL	Touch controller SCL	Clock out — add 4.7k pull-up to +3V3
GPIO3	TOUCH_IRQ	Touch controller IRQ	Touch interrupt IN
GPIO38	IR_TX	NPN transistor base for IR LED	PWM output for IR transmission
GPIO39	IR_RX	TSOP38238 IR receiver output	IR signal IN
GPIO7	BATT_ADC	Voltage divider from battery	Analog IN — battery level monitoring
GPIO19	USB_DN	USB-C connector D-	Fixed USB OTG pin — cannot change
GPIO20	USB_DP	USB-C connector D+	Fixed USB OTG pin — cannot change

TIP: GPIO33-37 are only available if you are NOT using octal PSRAM. Since Flipper Black uses external W25Q128 flash with no PSRAM, these GPIOs are free for SPI3.

The W25Q128 is the firmware storage chip. It connects to ESP32-S3 via SPI3. You placed SPI3 net labels on ESP32-S3 already. Now you place matching labels on the W25Q128 pins.

Pin connections

W25Q128 pin	Pin name	Net label	Notes
1	/CS	SPI3_CS_FLASH	Active LOW. Add 10k pull-up to +3V3 — keeps chip deselected at power-on
2	DO (MISO)	SPI3_MISO	Data out from flash to ESP32
3	/WP	Tie to +3V3 via 10k pull-up	Write protect — pull HIGH to disable write protection
4	GND	GND	Ground
5	DI (MOSI)	SPI3_MOSI	Data in from ESP32 to flash
6	CLK	SPI3_SCK	SPI clock from ESP32
7	/HOLD	Tie to +3V3 via 10k pull-up	Hold pin — pull HIGH to disable hold function
8	VCC	+3V3	100nF + 1uF decoupling caps to GND

TIP: After soldering, run 'esptool.py flash_id' to verify. Correct response: Manufacturer 0xEF, Device 0x4018. If you see 0xFF, there is a bad solder joint on one of the signal pins.

The CC1101 is the most complex chip in terms of external components. It has a digital side (SPI interface to ESP32-S3) and an RF side (antenna matching network). Do the digital side first, then the RF side.

3A — Digital side (SPI + power)

CC1101 pin	Pin name	Net label / connection	Notes
1	SCLK	SPI2_SCK	SPI clock from ESP32-S3
2	SO (GDO1)	SPI2_MISO	SPI data out to ESP32-S3
3	GDO2	CC1101_GDO2	Optional interrupt — connect to GPIO5. Add 10k pull-up to +3V3.
4	DVDD	Leave isolated — connect only to C51	DO NOT connect to your 3.3V rail. This is a regulated output FROM the chip. Connecting 3.3V here destroys it.
5	DCOUP	100nF cap to GND only	Decoupling for internal regulator. No connection to power rail.
6	GDO0 / CSn	GDO0: CC1101_GDO0 to GPIO4. CSn: SPI2_CS_CC1101	Two separate pins. GDO0 is the TX/RX state indicator. CSn is chip select.
7	GND	GND	Ground
8	XOSC_Q1	26MHz crystal pin 1 + 15pF cap to GND	CC1101 needs 26MHz — NOT 40MHz
9	AVDD	See power below	Analog VDD
10	XOSC_Q2	26MHz crystal pin 2 + 15pF cap to GND	
11	AVDD	See power below	
18	GND	GND	Die attach pad — connect to GND with multiple vias
19	GND	GND	
20	SI	SPI2_MOSI	SPI data in from ESP32-S3

Power pins — CC1101 has 4 AVDD pins

Pin	Name	Connection	Decoupling
9	AVDD	+3V3	100nF to GND
11	AVDD	+3V3	100nF to GND
14	AVDD	+3V3	100nF to GND
15	AVDD	+3V3	100nF + 1uF to GND (this is the main RF power pin)

Also add a ferrite bead or small inductor (R171 in the datasheet schematic) in series with the power supply trace feeding all AVDD pins. This filters switching noise from entering the chip. Use a 600 ohm @ 100MHz ferrite bead.

3B — RF matching network (copy from datasheet)

The RF matching network sits between the CC1101's RF_P and RF_N pins and the antenna. Its job is to match the chip's output impedance to 50 ohms (the antenna standard). Use values from TI's datasheet Table 5 and DN013. Values differ per frequency band.

Values for 433 MHz (ISM band — recommended for Flipper Black for maximum compatibility)

Component	Value	Location in circuit	Notes
L121	270nH	RF_P to balun	Multilayer OK for 433MHz
L122	270nH	RF_N to balun	Multilayer OK for 433MHz
L131	270nH	Balun inductor	
L132	270nH	Balun inductor	
C121	56pF	Matching cap	NPO/C0G dielectric — mandatory for RF
C122	56pF	Matching cap	NPO/C0G
C123	10pF	Matching cap	NPO/C0G
C124	12pF	Matching cap	NPO/C0G
C131	100nF	AVDD decoupling in RF section	+3V3 to GND, close to RF section
L123	39nH	Final matching inductor	Wire-wound for 433MHz
L124	39nH	Final matching inductor	Wire-wound
C125	100pF	DC blocking cap to antenna	Blocks DC, passes RF signal to antenna

Values for 868 MHz (if you choose this band instead)

Component	Value	Notes
L121	5.6nH	Wire-wound required for 868MHz — not multilayer
L122	5.6nH	Wire-wound required
C121	Not placed	Remove this component for 868MHz
C122	Not placed	Remove this component for 868MHz
C123	5.6pF	NPO/C0G
C124	2.7pF	NPO/C0G
L131	8.2nH	Wire-wound balun
L132	8.2nH	Wire-wound balun
C131	100nF	Decoupling
C125	100pF	DC blocking to antenna

TIP: Use TI SmartRF Studio 7 (free download at ti.com/tool/SMARTRFSTUDIO) to generate exact CC1101 register values for your chosen frequency. Copy the output directly into firmware. Never calculate manually.

CRITICAL: Never use multilayer inductors for 868MHz or 915MHz RF matching. Their Q factor is too low at high frequency. Use wire-wound inductors (Murata LQW series or similar). For 433MHz multilayer is acceptable.

SX1276 is your LoRa radio. It has a 28-pin QFN package. Connect SPI3 from ESP32-S3, plus dedicated interrupt and reset pins. The RF output goes through a simple matching network to a U.FL connector or PCB trace antenna.

4A — SPI and control pins

SX1276 pin	Pin name	Net label / connection	Notes
1	VDD_RXBPF	100nF to GND	Internal LNA supply — decouple to GND only
2	VDD_DCDC	Not connected (no connect marker)	Internal DC-DC — leave open unless using DC-DC mode
3	VDD	+ 3V3	100nF + 10uF to GND
5	RFIO	RF output — to matching network	50 ohm trace to matching network then U.FL
10	NSS	SPI3_CS_SX1276	Chip select. 10k pull-up to +3V3
11	SCK	SPI3_SCK	SPI clock from ESP32
12	MOSI	SPI3_MOSI	SPI data from ESP32
13	MISO	SPI3_MISO	SPI data to ESP32
14	RESET	SX1276_RST	Active LOW reset from GPIO47. 10k pull-up to +3V3.
15	DIO0	SX1276_DIO0	TX done/RX done interrupt to GPIO21. MUST connect — do not float.
16	DIO1	Optional — connect to free GPIO or add no-connect marker	RX timeout interrupt
17	DIO2	No connect marker	
18	DIO3	No connect marker	
19	DIO4	No connect marker	
20	DIO5	No connect marker	
21	GND	GND	Ground
28	XTAL_2	Crystal pin 2 + 10pF to GND	32MHz crystal for SX1276
27	XTAL_1	Crystal pin 1 + 10pF to GND	32MHz — different from ESP32's 40MHz

4B — RF matching network for 868 MHz

SX1276 RFIO pin connects to a simple pi-network matching circuit then to your antenna connector. The datasheet AN1200.13 provides these values.

Component	Value	Placement
L1 (RF choke)	47nH	Between VDD and RFIO — blocks DC from antenna path

Component	Value	Placement
C1	10pF NPO	RFIO to GND — part of pi matching
C2	Not placed (open)	Shunt cap — tune after testing
C3 (DC block)	100pF NPO	Series between matching network and antenna
Antenna connector	U.FL SMD	50 ohm impedance — route 50 ohm trace to here

TIP: Reference the Semtech SX1276 application note AN1200.13 (available at semtech.com) for the exact recommended matching network. For initial build just use the minimal circuit above — it works well enough to test range.

PN532 connects to SPI2 (same bus as CC1101, different CS pin). Interface mode is set permanently by two strapping pins using 0-ohm resistors — this is set before the board is made and cannot be changed in firmware.

5A — Interface mode selection (do this first)

PN532 has three interface modes. You must choose SPI mode. Set it by placing 0-ohm resistors on HSU0 and HSU1 pins. These resistors tie the pins to the correct voltage permanently.

Pin	Name	Connection for SPI mode	How to place it
HSU1	Interface select 1	Tie to GND via 0-ohm resistor	Place 0-ohm 0402 resistor between HSU1 and GND pad
HSU0	Interface select 0	Tie to +3V3 via 0-ohm resistor	Place 0-ohm 0402 resistor between HSU0 and +3V3 pad

WARNING: Do not hardwire HSU0/HSU1 directly to copper. Use 0-ohm resistors so you can change mode by reworking the resistors if needed. This was flagged in the improvement list.

5B — Pin connections

PN532 pin	Pin name	Net label / connection	Notes
1	TXD / MISO	SPI2_MISO	SPI data out to ESP32 in SPI mode
2	RXD / MOSI	SPI2_MOSI	SPI data in from ESP32 in SPI mode
3	SCK	SPI2_SCK	SPI clock
4	NSS	SPI2_CS_PN532	Chip select. 10k pull-up to +3V3.
8	IRQ	PN532_IRQ to GPIO6	Interrupt output — card detected. 10k pull-up to +3V3.
VDD	VDD	+3V3	100nF + 1uF to GND
RSTPDN	Reset	10k pull-up to +3V3. Optional GPIO for hard reset.	Active LOW reset
SVDD	SVDD	+3V3	100nF to GND — separate supply for RFID analog section
GND	GND	GND	

5C — NFC antenna coil

PN532 needs an external PCB coil antenna. This is a rectangular spiral trace on your PCB, not a component you buy. The datasheet application note AN10609 gives the exact coil dimensions and tuning capacitor values. Draw it as a symbol in KiCad and connect pins LXIN and LXOUT to it via matching capacitors.

PN532 pin	Connect to
LXIN	NFC antenna coil pin 1 via tuning cap (C = 22pF typical, tune per AN10609)
LXOUT	NFC antenna coil pin 2 via tuning cap

The GPS subsystem has three parts that must be designed together: the MAX-M8Q GPS chip (runs at 1.8V), the TLV70018 LDO (generates the 1.8V rail), and the TXB0102 level shifter (converts 1.8V UART to 3.3V for ESP32-S3). Do all three on the same schematic sheet.

6A — TLV70018 (1.8V LDO) — place this first

TLV70018 pin	Connection	Value
IN	+3V3 power symbol	10uF + 100nF to GND on input
OUT	+1V8 power symbol — new power rail	10uF + 100nF to GND on output
EN	Tie to IN (+3V3) via 100k pull-up	Always enabled — or connect to ESP32 GPIO for software control
GND	GND	

CRITICAL: MAX-M8Q absolute maximum voltage is 1.9V. If you accidentally connect +3V3 to its VCC pin it is destroyed instantly and permanently. Label the 1.8V rail clearly as +1V8 so it cannot be confused with +3V3.

6B — MAX-M8Q GPS pin connections

MAX-M8Q pin	Pin name	Connection	Notes
VCC	+1V8 rail	100nF + 1uF to GND — from TLV70018 output	NOT 3.3V — 1.8V only
VCC_IO	+1V8 rail	100nF to GND separately	IO voltage — must match UART logic level
GND	GND	Multiple vias — short path	
V_BCKP	Supercap (100mF) or coin cell (3V)	Maintains almanac between power cycles	Without this every startup is a 60s cold start. With it: 1s warm start.
TXD1	GPS_TX_1V8 net label	Goes to TXB0102 A1 pin (1.8V side)	GPS sends NMEA sentences here
RXD1	GPS_RX_1V8 net label	Goes to TXB0102 A2 pin (1.8V side)	GPS receives UBX config commands here
TIMEPULSE	Optional — no connect or GPIO	1 pulse per second when satellite locked	
RESET_N	10k pull-up to +1V8	Add optional GPIO for software reset	Active LOW
RF_IN	50 ohm trace to U.FL connector	Passive ceramic patch antenna	Keep trace short. No ground pour under antenna area.

6C — TXB0102 level shifter pin connections

TXB0102 translates between 1.8V (GPS side) and 3.3V (ESP32 side). It has an A-side (1.8V) and a B-side (3.3V). OE pin must be tied HIGH to 3.3V — if it floats the chip outputs nothing.

TXB0102 pin	Connection	Notes
VCCA	+1V8	1.8V supply for GPS side. 100nF to GND.
VCCB	+3V3	3.3V supply for ESP32 side. 100nF to GND.
OE	Tie directly to VCCB (+3V3)	CRITICAL — must be HIGH. Floating OE = no data passes through.
A1 (1.8V side)	GPS_TX_1V8 (from MAX-M8Q TXD1)	GPS sends here
A2 (1.8V side)	GPS_RX_1V8 (to MAX-M8Q RXD1)	GPS receives here
B1 (3.3V side)	ESP_UART_RX (GPIO44 net label)	Translated signal to ESP32
B2 (3.3V side)	ESP_UART_TX (GPIO43 net label)	Translated signal from ESP32
GND	GND	

WARNING: GPIO43 on ESP32-S3 is the default boot serial console TX pin. Add a 100 ohm series resistor between GPIO43 and the TXB0102 B2 pin. This prevents the boot console from fighting the GPS UART at startup.

The display uses SPI for pixel data and I2C for touch input. The exact pin names depend on your specific display module. Common controllers are ILI9488 or ST7796S for the display, FT6336 or CST816 for touch. Check your display's datasheet.

Signal	Net label	ESP32 GPIO	Notes
Display SPI MOSI	DISP_MOSI	GPIO15	Data from ESP32 to display
Display SPI SCK	DISP_SCK	GPIO16	Clock
Display CS	DISP_CS	GPIO17	Chip select. 10k pull-up to +3V3.
Display DC	DISP_DC	GPIO18	Data/Command select — HIGH=data, LOW=command
Display RST	DISP_RST	GPIO8	Active LOW reset. 10k pull-up to +3V3.
Display VCC	+3V3	—	100nF + 10uF to GND
Touch SDA	I2C_SDA	GPIO1	4.7k pull-up to +3V3
Touch SCL	I2C_SCL	GPIO2	4.7k pull-up to +3V3
Touch IRQ	TOUCH_IRQ	GPIO3	Interrupt — touch detected. 10k pull-up to +3V3.
Touch VCC	+3V3	—	100nF to GND
Backlight	Connect via NPN transistor or MOSFET to GPIO	—	PWM brightness control via transistor — do not connect LED directly to GPIO

IR TX uses a high-power IR LED driven by an NPN transistor (2N2222 or MMBT2222 SMD). The transistor is needed because GPIO can only source 40mA — the IR LED needs more for good range. IR RX uses a demodulating receiver IC (TSOP38238) that handles filtering and demodulation automatically.

Component	Pin	Net label / connection	Notes
NPN transistor (MMBT2222)	Base	IR_TX (GPIO38) via 100 ohm resistor	Resistor limits base current
NPN transistor	Collector	IR LED array anode	Up to 3 IR LEDs in series
NPN transistor	Emitter	GND	
IR LED array	Anode	Transistor collector	High-power 940nm IR LEDs (TSAL6400 or similar)
IR LED array	Cathode	GND	33 ohm current limiting resistor in series
TSOP38238 receiver	OUT	IR_RX (GPIO39)	Output is demodulated IR signal
TSOP38238	VCC	+3V3	100nF + 10uF to GND — datasheet requires good decoupling here
TSOP38238	GND	GND	

The power system has four parts: the TP4056 charges the battery from USB-C. The DW01A protects against overcharge, overdischarge, and short circuit. The AP2112K converts battery voltage to clean 3.3V. Design these in order.

9A — TP4056 battery charger

TP4056 pin	Connection	Notes
VCC	VBUS (+5V from USB-C)	100nF to GND
GND	GND	
PROG	Resistor to GND	Sets charge current. 1.2kohm = 1A charge current.
TEMP	10k pull-up to VCC	No thermistor — just pull up
CHRG	LED + resistor to GND (optional)	Charge indicator LED — shows when charging
STDBY	LED + resistor to GND (optional)	Standby indicator LED — shows charge complete
BAT	Battery positive via DW01A	Output to battery — goes through DW01A first

9B — DW01A protection IC + FS8205 dual MOSFET

DW01A works together with a dual N-channel MOSFET (FS8205A) to physically disconnect the battery under fault conditions. They must be used together.

DW01A pin	Connection
VDD	Battery positive (after TP4056 BAT pin)
GND / CS	Battery negative + current sense resistor to GND
OC	FS8205A gate 1 — controls overdischarge cutoff
OD	FS8205A gate 2 — controls overcharge cutoff

9C — AP2112K-3.3 LDO regulator

AP2112K pin	Connection	Notes
VIN	Battery positive (from DW01A protected output)	1uF + 10uF to GND on input
GND	GND	
EN	Tie to VIN via 100k pull-up	Always enabled. Or connect to GPIO for software power control.
VOUT	+3V3 power rail	10uF + 100nF to GND on output. This feeds every chip on the board.
VOUT (bulk)	Add 100uF electrolytic to GND	Handles large transient current spikes from WiFi and RF TX

9D — Battery connector

Connector pin	Connection
Pin 1 (positive)	TP4056 BAT and DW01A VDD
Pin 2 (negative)	GND via DW01A / FS8205A current sense path

TIP: Use a JST-PH 2.0mm 2-pin connector for the battery. This is the standard for 18650 and LiPo cells. Orient it on the PCB so polarity is obvious — mark + and - on the silkscreen.

10A — BOOT and EN buttons

Both buttons must be accessible from outside the aluminum enclosure without disassembly. Route them to pinhole buttons in the enclosure wall or to a header accessible from outside.

Button	Connects between	Net label	Notes
BOOT button	GPIO0 and GND	BOOT_BTN	GPIO0 already has 10k pull-up from Chip 1 strapping pins
EN (reset) button	CHIP_PU and GND	EN_BTN	CHIP_PU already has 10k pull-up and 1uF RC filter from Chip 1

10B — Test points (mandatory for debugging raw chips)

Test points are bare 1mm copper pads with no solder mask. You probe them with a scope or multimeter. Add one on every critical net. Raw QFN chips at 0.5mm pitch are impossible to probe directly.

Test point	Net to connect	Why
TP1	SPI2_MOSI	Debug CC1101 and PN532 data
TP2	SPI2_SCK	Debug SPI2 clock
TP3	SPI2_MISO	Debug SPI2 response
TP4	SPI3_MOSI	Debug SX1276 and flash data
TP5	SPI3_SCK	Debug SPI3 clock
TP6	ESP_UART_TX	Debug GPS UART output
TP7	ESP_UART_RX	Debug GPS NMEA input
TP8	+3V3	Probe main power rail
TP9	+1V8	Probe GPS power rail
TP10	GND	Reference probe point

10C — USB-C connector

USB-C pin	Connection	Notes
VBUS	VBUS rail to TP4056 VCC	Add ESD protection diode (USBLC6-2SC6) on VBUS — protects from USB spikes
D+	USB_DP (GPIO20)	Fixed USB OTG pin on ESP32-S3
D-	USB_DN (GPIO19)	Fixed USB OTG pin on ESP32-S3
CC1, CC2	5.1k resistors to GND each	Required for USB-C power negotiation — without these chargers may not provide power
GND	GND	Shield and signal ground

Final checklist before moving to PCB layout

Check	What to verify
ERC clean	Run Inspect > Electrical Rules Check. Zero errors. All warnings reviewed and justified.
Every VCC pin has decoupling	Open each chip symbol. Confirm 100nF on every VCC pin.
Every CS pin has pull-up	10k pull-up to +3V3 on: CC1101 CSn, PN532 NSS, SX1276 NSS, W25Q128 /CS
Strapping pins correct	GPIO0 pull-up + button. GPIO45 pull-down. GPIO46 pull-down. EN pull-up + RC + button.
PN532 HSU pins set	HSU1 to GND via 0-ohm. HSU0 to +3V3 via 0-ohm.
TXB0102 OE pin HIGH	OE tied directly to VCCB (3.3V). Not floating.
MAX-M8Q on 1.8V only	+1V8 symbol on MAX-M8Q VCC and VCC_IO. Never +3V3.
Crystal values correct	ESP32-S3: 40MHz. CC1101: 26MHz. SX1276: 32MHz. All different.
No connect markers placed	Every intentionally unused pin has an X marker
Net labels consistent	SPI2_MOSI on ESP32-S3 GPIO11 matches SPI2_MOSI on CC1101 SI exactly — same spelling, same case
All footprints assigned	Tools > Assign Footprints — no components with missing footprint